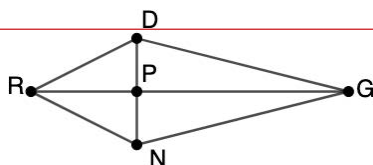


Geometry _____ Name _____
 Unit 8 _____ Date _____
 Lesson 4 Homework _____ Period _____

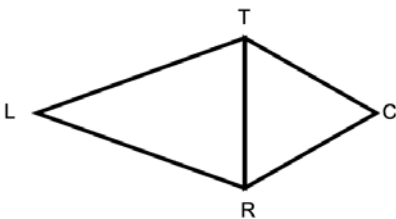
Kite $RDGN$ is shown with diagonals $\overline{RG}$ and $\overline{DN}$ intersecting at point P where $DP = 13$ and $RP = 15$.



Commented [DB1]: <https://www.geogebra.org/calculator/c6hkhhw>

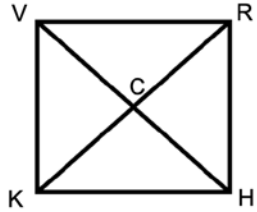
1. What is the length of $\overline{RD}$?
2. Given $PG = 22$. What is the length of $\overline{NG}$?
3. What is the perimeter of $RDGN$?

TCRL is a kite as shown where $m\angle CTR = (7x - 12)^\circ$, $m\angle TCR = (8x + 32)^\circ$, $m\angle LRT = (9y + 40)^\circ$, and $m\angle TLR = (4y)^\circ$.



4. What is the value of x ? Round your answer to the nearest hundredth.
5. What is the value of y ? Round your answer to the nearest hundredth.
6. What is the measure of $\angle CTL$?

Quadrilateral $VRHK$ is shown with diagonals $\overline{VH}$ and $\overline{RK}$ intersecting at point C .



7. What must be true for $VRHK$ to be a square?
8. Given $VRHK$ is a square with $VC = 6z$, and $VR = 12$. What is the value of z ? Leave your answer in radical form.
9. What is the length of $\overline{RK}$, in simplest radical form?
10. What is the length of $\overline{VK}$, in simplest radical form?